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Delayed acceptance particle MCMC for exact inference in stochastic kinetic models

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Abstract Recently-proposed particle MCMC methods provide a flexible way of performing Bayesian inference for parameters governing stochastic kinetic models defined as Markov (jump) processes (MJPs). Each iteration of the scheme requires an estimate of the marginal likelihood calculated from the output of a sequential Monte Carlo scheme (also known as a particle filter). Consequently, the method can be extremely computationally intensive. We therefore aim to avoid most instances of the expensive likelihood calculation through use of a fast approximation. We consider two approximations: the chemical Langevin equation diffusion approximation (CLE) and the linear noise approximation (LNA). Either an estimate of the marginal likelihood under the CLE, or the tractable marginal likelihood under the LNA can be used to calculate a first step acceptance probability. Only if a proposal is accepted under the approximation do we then run a sequential Monte Carlo scheme to compute an estimate of the marginal likelihood under the true MJP and construct a second stage acceptance probability that permits exact (simulation based) inference for the MJP. We therefore avoid expensive calculations for proposals that are likely to be rejected. We illustrate the method by considering inference for parameters governing a Lotka–Volterra system, a model of gene expression and a simple epidemic process.

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1 Introduction

Stochastic kinetic models describe a probabilistic mechanism for the joint evolution of species in a dynamical system. They can be used to model a wide variety of real-world phenomena and are increasingly applied in computational systems biology (Kitano 2002), motivated by a need for models that incorporate intrinsic stochasticity (Elowitz et al. 2002; Swain et al. 2002; Wilkinson 2009). Other areas of application include (but are not limited to) predator-prey population models (Boys et al. 2008; Ferm et al. 2008; Golightly and Wilkinson 2011) and epidemic models (O'Neill and Roberts 1999; Boys and Giles 2007; Ball and Neal 2008; Jewell et al. 2009). Underpinned by a reaction network in which reaction events change species numbers by an integer amount, a stochastic kinetic model is most naturally represented by a continuous time Markov jump process (MJP). Our goal is to perform inference for the rate constants that govern the MJP using time course data that may be incomplete and/or subject to measurement error.

Exact (simulation based) Bayesian inference for the MJP was the subject of Boys et al. (2008). The authors proposed two MCMC schemes that targeted the joint posterior of the rate constants and latent reaction events but found the statistical efficiency of their method to be relatively poor. It was shown in Golightly and Wilkinson (2011) how a recently proposed particle MCMC algorithm (Andrieu et al. 2010) can be applied to this class of models. In particular, the particle marginal Metropolis-Hastings (PMMH) scheme allows a joint update of the rate constants and (latent) process which

can alleviate common mixing problems when sampling high dimensional target densities that may exhibit strong correlations. The proposal mechanism involves drawing a new parameter value from an arbitrary proposal kernel and drawing new values of each latent state from a sequential Monte Carlo (SMC) approximation to the distribution of latent states conditional on the proposed new parameter value. The acceptance probability requires computation of a realisation of an unbiased estimator of marginal likelihood which can be readily obtained from the output of the SMC scheme. Consequently, at each iteration of the MH scheme, an SMC algorithm must be implemented. The method can be extremely computationally intensive, as the SMC algorithm typically must generate many realisations of the MJP, with each realisation obtained from an algorithm such as the stochastic simulation algorithm (SSA) of Gillespie (1977). By using a computationally cheaper approximation to the marginal likelihood we avoid running the computationally more expensive SMC algorithm at most iterations of the MH scheme, but we still maintain the posterior under the MJP as the target distribution of the MH scheme.

The simplest approximation of the MJP is the macroscopic rate equation (MRE) which ignores the discreteness and stochasticity of the MJP by modelling specie dynamics with a set of coupled ordinary differential equations (van Kampen 2001). The diffusion approximation or chemical Langevin equation (CLE) (Gillespie 2000) on the other hand, ignores discreteness but not stochasticity by modelling the reaction network with a set of coupled stochastic differential equations (SDEs). Whilst inference for the parameters governing nonlinear multivariate SDEs is possible (Golightly and Wilkinson 2008), the marginal likelihood under this model is intractable. Despite this, Golightly and Wilkinson (2011) show that inference is possible under this model using a PMMH algorithm, and this approach can result in computational savings when compared to a similar scheme targeting the posterior under the MJP.

Further computational savings can be made by considering a linear noise approximation (LNA) (van Kampen 2001; Komorowski et al. 2009; Fearnhead et al. 2014) which is given by the MRE plus a stochastic term accounting for random fluctuations about the MRE. Under the LNA the latent process follows a multivariate Gaussian distribution and, under an assumption of Gaussian measurement error, the marginal likelihood is tractable.

Christen and Fox (2005) describe a delayed-acceptance Metropolis-Hastings scheme in which the single MH accept-reject step is replaced by an initial 'screening' stage which substitutes a computationally cheap approximate posterior for the true posterior in the MH acceptance probability formula, but then adds a second accept-reject stage which ensures that detailed balance is still satisfied with respect to the true posterior. This second, computationally expensive, stage is only applied to proposals which pass the first stage.

Our novel contribution is to exploit the tractability of the LNA by proposing a particle analogue of this scheme for performing exact, simulation based inference for the MJP parameters. Essentially, to avoid calculating an estimate of marginal likelihood under the MJP for proposals that are likely to be rejected, proposed parameter draws are initially screened using a computationally cheap approximation to the posterior, such as that based on the marginal likelihood computed under the LNA. A related approach has been proposed independently by Smith (2011) for performing inference for the parameters governing nonlinear, discrete time economic models. A simple stochastic volatility model and a Real Business Cycle model are considered, with approximations based on a linear Gaussian state space model and an unscented Kalman filter used in a preliminary screening step. Unlike Smith (2011), we also consider a scenario in which the marginal likelihood under the approximation is intractable, but can be estimated cheaply (relative to the same calculation under the MJP) using a particle filter. Use of the CLE in the preliminary screening step falls into this category. In both cases, we show that the resulting MCMC scheme targets the correct marginal, that is, the marginal parameter posterior under the MJP. The proposed methods can in principle be applied to any Markov jump process.

The remainder of this paper is organised as follows. In Sect. 2 we describe the Markov jump process model and associated inference problem. The CLE and LNA are briefly reviewed. We describe the PMMH algorithm in Sect. 3.1 before considering a modification to allow delayed acceptance in Sect. 3.3. We apply the method to a Lotka–Volterra system, a model of gene expression and a simple epidemic process in Sect. 4. Conclusions are drawn in Sect. 5.

2 Stochastic kinetic models

Consider a reaction network involving u species $\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_u$ and v reactions $\mathcal{R}_1, \mathcal{R}_2, \ldots, \mathcal{R}_v$, with reaction $\mathcal{R}_i$ given by

$$\begin{aligned}\mathcal{R}_i : \quad & p_{i1}\mathcal{X}_1 + p_{i2}\mathcal{X}_2 + \cdots + p_{iu}\mathcal{X}_u \\ \longrightarrow\ & q_{i1}\mathcal{X}_1 + q_{i2}\mathcal{X}_2 + \cdots + q_{iu}\mathcal{X}_u\end{aligned}$$

where the stoichiometric coefficients p_{ij} and q_{ij} are non-negative integers. Let $X_{j,t}$ denote the number of specie $\mathcal{X}_j$ at time t, and let X_t be the u-vector $X_t = (X_{1,t}, X_{2,t}, \ldots, X_{u,t})'$. The $v \times u$ matrix P consists of the coefficients p_{ij}, and Q is defined similarly. The $u \times v$ stoichiometry matrix S is defined by

$$S = (Q - P)'$$

and encodes important structural information about the reaction network. In particular, if ΔR is a v-vector containing the number of reaction events of each type in a given time interval, then the system state should be updated by ΔX, where

$$\Delta X = S\Delta R.$$

Each reaction $\mathcal{R}_i$ is assumed to have an associated rate constant, c_i, and a propensity function, $h_i(X_t, c_i)$ giving the overall hazard of a type i reaction occurring. That is, we model the system as a Markov jump process, and for an infinitesimal time increment dt, the probability of a type i reaction occurring in the time interval $(t, t + dt]$ is $h_i(X_t, c_i)dt$. In many examples (such as those considered in Sects. 4.1 and 4.3) the form of $h_i(X_t, c_i)$ can be thought of as arising naturally from the interactions between components of a well-mixed population, such as reactants in a well-stirred container at constant temperature. This leads to a mass action kinetic rate law (Gillespie 1992), under which the hazard function for a particular reaction of type i takes the form

$$h_i(X_t, c_i) = c_i \prod_{j=1}^{u} \binom{X_{j,t}}{p_{ij}}.$$

Let $c = (c_1, c_2, \ldots, c_v)'$ and $h(X_t, c) = (h_1(X_t, c_1), h_2(X_t, c_2), \ldots, h_v(X_t, c_v))'$. Values for c and the initial system state $X_0 = x_0$ complete specification of the Markov process. Although this process is rarely analytically tractable for interesting models, it is straightforward to forward-simulate exact realisations of this Markov process using a discrete event simulation method. This is due to the fact that if the current time and state of the system are t and X_t respectively, then the time to the next event will be exponential with rate parameter

$$h_0(X_t, c) = \sum_{i=1}^{v} h_i(X_t, c_i),$$

and the event will be a reaction of type $\mathcal{R}_i$ with probability $h_i(X_t, c_i)/h_0(X_t, c)$ independently of the waiting time. Forward simulation of process realisations in this way is typically referred to as the stochastic simulation algorithm (Gillespie 1977). See Wilkinson (2012) for further background on stochastic kinetic modelling.

2.1 Chemical Langevin equation

We present here an informal intuitive construction of the chemical Langevin equation (CLE), and refer the reader to Gillespie (2000) for further details.

Consider an infinitesimal time interval, $(t, t + dt]$. Over this time, the reaction hazards will remain constant almost surely. The occurrence of reaction events can therefore be regarded as the occurrence of events of a Poisson process with independent realisations for each reaction type. Therefore, if we write dR_t for the v-vector of the number of reaction events of each type in the infinitesimal time increment, it is clear that the elements are independent of one another and that the ith element is a $Po(h_i(X_t, c_i)dt)$ random quantity. From this we have that $\mathrm{E}(dR_t) = h(X_t, c)dt$ and $\mathrm{Var}(dR_t) = \mathrm{diag}\{h(X_t, c)\}dt$. It is therefore clear that

$$dR_t = h(X_t, c)dt + \mathrm{diag}\left\{\sqrt{h(X_t, c)}\right\} dW_t$$

is the Itô stochastic differential equation (SDE) which has the same infinitesimal mean and variance as the true Markov jump process (where dW_t is the increment of a v-dimensional Brownian motion). Now since $dX_t = SdR_t$, we obtain

$$dX_t = S\,h(X_t, c)dt + \sqrt{S\mathrm{diag}\{h(X_t, c)\}S'}dW_t, \tag{1}$$

where now X_t and W_t are both u-vectors. Equation (1) is the SDE most commonly referred to as the chemical Langevin equation or diffusion approximation, and represents the diffusion process which most closely matches the dynamics of the associated Markov jump process, and can be shown to approximate the stochastic kinetic model increasingly well in high concentration scenarios (Gillespie 2000). Note that in the absence of an analytic solution to (1), a numerical solution can be constructed. For example, the Euler-Maruyama approximation is

$$\begin{aligned}\Delta X_t &\equiv X_{t+\Delta t} - X_t \\ &= S\,h(X_t, c)\Delta t + \sqrt{S\mathrm{diag}\{h(X_t, c)\}S'}\Delta W_t\end{aligned} \tag{2}$$

where ΔW_t is a mean zero Normal random vector with variance matrix $\mathrm{diag}\{\Delta t\}$.

We require a computationally efficient approximation to the Markov jump process for use in a delayed acceptance particle MCMC scheme (described in Sect. 3.3). Performing exact (simulation based) inference for the diffusion approximation has been the focus of Golightly and Wilkinson (2005), Purutcuoglu and Wit (2007), and Golightly and Wilkinson (2011) among others. Although the latter find that a particle MCMC scheme based on the CLE can be more computationally efficient than a similar scheme that works with the Markov jump process directly, calculation of an estimate of marginal likelihood under the CLE (as is necessary at every iteration of a particle MCMC scheme) can be computationally expensive. To facilitate greater computational savings, we therefore also consider a linear noise approximation (LNA) (van Kampen 2001; Komorowski et al. 2009; Fearnhead et al. 2014; Stathopoulos and Girolami 2013) which generally possesses a greater degree of numerical and analytic tractability than the CLE (Wilkinson 2012). This is the subject of the next section.

2.2 Linear noise approximation

The LNA was first considered as a functional central limit law for density dependent processes by Kurtz (1970) and can be derived in a number of more or less formal ways. For example, Komorowski et al. (2009) (and see also Elf and Ehrenberg (2003)) derive the LNA by approximating the forward Kolmogorov equation (satisfied by the transition rate of the MJP) through a Taylor series expansion. We eschew this approach in favour of an informal derivation following that of Fearnhead et al. (2014) and we refer the reader to the references therein for a more detailed discussion. In what follows we calculate the LNA for a general SDE before formulating it as an approximation to the CLE.

Consider now a general SDE satisfied by a process $\{X_t, t \geq 0\}$ of the form

$$dX_t = \alpha(X_t)dt + \epsilon\beta(X_t)dW_t \tag{3}$$

where $\epsilon << 1$. Partition X_t into a deterministic path z_t and a residual stochastic process M_t and let z_t be the solution to

$$\frac{dz_t}{dt} = \alpha(z_t). \tag{4}$$

We assume that $||X_t - z_t||$ is $O(\epsilon)$ over a time interval of interest and substitute $X_t = z_t + \epsilon M_t$ into Eq. (3) to give

$$d(z_t + \epsilon M_t) = \alpha(z_t + \epsilon M_t)dt + \epsilon\beta(z_t + \epsilon M_t)dW_t.$$

We then Taylor expand $\alpha(\cdot)$ and $\beta(\cdot)$ about z_t and collect terms of $O(\epsilon)$ to give the SDE satisfied by M_t as

$$dM_t = F_t M_t dt + \beta(z_t)dW_t \tag{5}$$

where F_t is the Jacobian matrix with (i, j)th element $\partial\alpha_i(z_t)/\partial z_{j,t}$ and $\alpha_i(z_t)$ refers to the ith element of $\alpha(z_t)$. Note that we use ϵ to indicate that the stochastic term in (3) is small and, essentially, that the drift term dominates the diffusion coefficient. However ϵ plays no role in the evolution equations, (4) and (5). Without loss of generality, therefore, we simplify the exposition by setting $\epsilon = 1$. To further simplify the notation we also drop the explicit dependence of the hazard function on c, and of the mean and variance of M_t on both c and z_t.

For the CLE, we have

$$\alpha(X_t) = S\,h(X_t), \qquad \beta(X_t) = \sqrt{S\text{diag}\{h(X_t)\}S'}.$$

The linear noise approximation of the CLE is therefore defined through

$$\frac{dz_t}{dt} = Sh(z_t) \tag{6}$$

and

$$dM_t = F_t M_t dt + \sqrt{S\text{diag}\{h(z_t)\}S'}dW_t \tag{7}$$

where F_t has (i, j)th element $S\partial h_i(z_t)/\partial z_{j,t}$.

For fixed or Gaussian initial conditions, that is $M_{t_1} \sim \text{N}(m_{t_1}, V_{t_1})$, the SDE in (7) can be solved explicitly to give

$$(M_t|c) \sim \text{N}\,(m_t\,,\ V_t) \tag{8}$$

where m_t is the solution to the deterministic ordinary differential equation (ODE)

$$\frac{dm_t}{dt} = F_t m_t \tag{9}$$

and similarly

$$\frac{dV_t}{dt} = V_t F_t' + S\text{diag}\{h(z_t)\}S' + F_t V_t\,. \tag{10}$$

Hence, the solution of Eq. (7) requires the solution of a system of coupled ODEs; in the absence of an analytic solution to these equations, a numerical solution can be used. The approximating distribution of X_t can then be found as

$$(X_t|c) \sim \text{N}\,(z_t + m_t\,,\ V_t)\,. \tag{11}$$

3 Inference

We now consider the task of performing inference for the rate constants governing the Markov jump process. First, let us augment the rate vector c to include any additional parameters that arise from the observation process and assign to it a prior density, $p(c)$. Suppose that the MJP $\mathbf{X} = \{X_t \mid 1 \leq t \leq T\}$ is not observed directly, but (perhaps partial) observations (on a regular grid) $\mathbf{y} = \{y_t \mid t = 1, 2, \ldots, T\}$ are available and assumed conditionally independent (given $\mathbf{X}$) with conditional probability distribution $p(y_t|x_t, c)$.

In this work, we consider Bayesian inference for c via the marginal posterior density

$$p(c|\mathbf{y}) = \int p(c, \mathbf{x}|\mathbf{y})\,\mathrm{d}\mathbf{x} \tag{12}$$

where

$$p(c, \mathbf{x}|\mathbf{y}) \propto p(c)\,p(\mathbf{x}|c)\prod_{t=1}^{T} p(y_t|x_t, c)$$

and $p(\mathbf{x}|c)$ is the probability of the Markov jump process. Since the posterior in (12) will typically be unavailable in closed form, samples must usually be generated through a suitable MCMC scheme.

In what follows, for simplicity, we assume that the initial value of the MJP, $X_1 = x_1$, is a known fixed quantity, and we take $z_1 = x_1$ so that m_1 is the length-u zero vector and V_1 is the $u \times u$ zero matrix. If X_1 were unknown then it could be assigned a prior and treated as an additional parameter in the augmented rate vector.

3.1 Particle marginal Metropolis-Hastings

We consider the special case of the particle marginal Metropolis-Hastings (PMMH) scheme of Andrieu et al. (2009, 2010) in which only samples from the marginal parameter posterior are required. Noting the standard decomposition $p(c|\mathbf{y}) \propto p(\mathbf{y}|c)p(c)$, we run a Metropolis-Hastings (MH) scheme with proposal kernel $q(c^*|c)$ and accept a move from c to c^* with probability

$$\min\left\{1\,,\ \frac{\widehat{p}(\mathbf{y}|c^\star)\,p(c^\star)}{\widehat{p}(\mathbf{y}|c)\,p(c)} \times \frac{q(c|c^\star)}{q(c^\star|c)}\right\} \tag{13}$$

where $\widehat{p}(\mathbf{y}|c)$ is a sequential Monte Carlo (SMC) or 'particle filter' estimate of the intractable marginal likelihood term $p(\mathbf{y}|c)$. The PMMH scheme as described here is an example of a pseudo-marginal Metropolis-Hastings scheme (Beaumont 2003; Andrieu and Roberts 2009) and provided that $\widehat{p}(\mathbf{y}|c)$ is unbiased (or has a constant multiplicative bias that does not depend on c), it is possible to verify that the method targets the marginal $p(c|\mathbf{y})$. Let u denote all random variables generated by the SMC algorithm and write the SMC estimate of marginal likelihood as $\widehat{p}(\mathbf{y}|c) = \widehat{p}(\mathbf{y}|c, u)$. Augmenting the state space of the chain to include u, it is straightforward to rewrite the acceptance ratio in (13) to find that the chain targets the joint density

$$\widehat{p}(c, u|\mathbf{y}) \propto \widehat{p}(\mathbf{y}|c, u)\,p(u|c)\,p(c)\,.$$

Marginalising over u then gives

$$\int \widehat{p}(c, u|\mathbf{y})du \propto p(c) \int \widehat{p}(\mathbf{y}|c, u)\,p(u|c)du$$
$$\propto p(c)\,p(\mathbf{y}|c).$$

The key insight here is that the SMC scheme can be constructed to give an unbiased estimate of the marginal likelihood $p(\mathbf{y}|c)$ under some fairly mild conditions involving the resampling scheme (Del Moral 2004). The scheme therefore targets the correct marginal $p(c|\mathbf{y})$. Although interest here is in the marginal $p(c|\mathbf{y})$ the PMMH scheme can be used to sample the joint density $p(c, \mathbf{x}|\mathbf{y})$. At each step of the algorithm, a new path $\mathbf{x}^*$ is proposed from an SMC approximation of $p(\mathbf{x}^*|\mathbf{y}, c^*)$. The acceptance probability is as in (13). For further details, we refer the reader to Andrieu et al. (2010). The (special case of the) PMMH algorithm and details of the SMC scheme that we use are given in the sections "PMMH scheme" and "SMC scheme" of Appendix.

3.2 Inference using the CLE and LNA

Although the marginal likelihood under the CLE is intractable, a PMMH scheme can be implemented to perform inference for this model. In the simplest version of the scheme, we replace draws of the MJP in step 2(a) of the SMC scheme with draws of a numerical solution of the CLE, for example, using the Euler-Maruyama approximation. This is the focus of Golightly and Wilkinson (2011) and further details can be found therein.

Under the LNA, the marginal likelihood is tractable for additive Gaussian observation regimes. This tractability has been exploited for the purposes of parameter inference by Komorowski et al. (2009), Fearnhead et al. (2014) and Stathopoulos and Girolami (2013). In Komorowski et al. (2009) and Stathopoulos and Girolami (2013), the LNA is applied over the entire time interval for which observations are available. In particular, the ODE component of the LNA is solved once over the whole time-course for a given initial condition. As discussed in Fearnhead et al. (2014), this can lead to a poor approximation to the distribution of X_t as t gets large, due to the mismatch between the stochastic and ODE solution. We therefore adopt the approach proposed in Fearnhead et al. (2014) and restart the LNA at each observation time t, initialising z_t to the posterior mean of X_t given all observations up to time t. The algorithm for constructing the marginal likelihood under an additive Gaussian observation regime using this approach is given in the section " Marginal likelihood under the linear noise approximation" of Appendix. Use of a Gaussian observation model is likely to be unsatisfactory in some scenarios. For example, in Sect. 4.1 we consider observations with a Poisson distribution, the mean of which is the value of the true process. Nonetheless, we may still use the LNA to obtain a tractable approximation to the marginal likelihood under the true MJP. We approximate the observation density $p(y_t|x_t)$ by a Gaussian density with mean and variance given by the ODE solution (6). That is, we apply the algorithm in the section " Marginal likelihood under the linear noise approximation" of Appendix with Σ replaced by a diagonal matrix containing the components of z_t for which observations are made. This tractable approximation can then be used in the delayed acceptance scheme.

3.3 Delayed acceptance particle marginal Metropolis-Hastings

In order to improve the efficiency of the PMMH algorithm for the MJP we aim to limit the number of runs of the computationally expensive SMC scheme for the MJP. Ideally we want to run the SMC scheme only for parameter values which

Algorithm 1 Delayed acceptance PMMH (daPMMH)

1. Initialisation, $i = 0$,
   (a) set $c^{(0)}$ arbitrarily,
   (b) run a particle filter targeting $p(\mathbf{x}|\mathbf{y}, c^{(0)})$, and let $\widehat{p}(\mathbf{y}|c^{(0)})$ denote the marginal likelihood estimate,
   (c) run a particle filter targeting $p_a(\mathbf{x}|\mathbf{y}, c^{(0)})$, and let $\widehat{p}_a(\mathbf{y}|c^{(0)})$ denote the marginal likelihood estimate under the approximate model.
2. For iteration $i \geq 1$,
   (a) sample $c^* \sim q(\cdot|c^{(i-1)})$,
   (b) **Stage 1**
      (i) run a particle filter targeting $p_a(\mathbf{x}|\mathbf{y}, c^*)$, and let $\widehat{p}_a(\mathbf{y}|c^*)$ denote the marginal likelihood estimate under the approximate model,
      (ii) with probability

$$\alpha_1(c^{(i-1)}, c^*) = \min\left\{1, \frac{\widehat{p}_a(\mathbf{y}|c^*)p(c^*)}{\widehat{p}_a(\mathbf{y}|c^{(i-1)})p(c^{(i-1)})}\frac{q(c^{(i-1)}|c^*)}{q(c^*|c^{(i-1)})}\right\}, \tag{14}$$

      run a particle filter targeting $p(\mathbf{x}|\mathbf{y}, c^*)$, let $\widehat{p}(\mathbf{y}|c^*)$ denote the marginal likelihood estimate and go to 2(c); otherwise, set $c^{(i)} = c^{(i-1)}$, $\widehat{p}(\mathbf{y}|c^{(i)}) = \widehat{p}(\mathbf{y}|c^{(i-1)})$, $\widehat{p}_a(\mathbf{y}|c^{(i)}) = \widehat{p}_a(\mathbf{y}|c^{(i-1)})$, increment i and return to 2(a).
   (c) **Stage 2**
   With probability

$$\alpha_2(c^{(i-1)}, c^*) = \min\left\{1, \frac{\widehat{p}(\mathbf{y}|c^*)p(c^*)}{\widehat{p}(\mathbf{y}|c^{(i-1)})p(c^{(i-1)})}\frac{\widehat{p}_a(\mathbf{y}|c^{(i-1)})p(c^{(i-1)})}{\widehat{p}_a(\mathbf{y}|c^*)p(c^*)}\right\} \tag{15}$$

   set $c^{(i)} = c^*$, $\widehat{p}(\mathbf{y}|c^{(i)}) = \widehat{p}(\mathbf{y}|c^*)$ and $\widehat{p}_a(\mathbf{y}|c^{(i)}) = \widehat{p}_a(\mathbf{y}|c^*)$ otherwise set $c^{(i)} = c^{(i-1)}$, $\widehat{p}(\mathbf{y}|c^{(i)}) = \widehat{p}(\mathbf{y}|c^{(i-1)})$ and $\widehat{p}_a(\mathbf{y}|c^{(i)}) = \widehat{p}_a(\mathbf{y}|c^{(i-1)})$. Increment i and return to 2(a).

are likely to lead to acceptance in the PMMH algorithm. We do this by choosing a particular proposal kernel in the section "PMMH scheme" of Appendix. This proposal kernel is based on a preliminary screening step involving an approximate model which is less computationally intensive than the MJP, such as the LNA or the CLE. In what follows, the CLE approximation refers to the Euler-Maruyama approximation in (2). Likewise, the LNA refers to the numerical solution of the ODEs in (6), (9) and (10). We note that the CLE or LNA are used only in the preliminary screening step and further approximation through use of a numerical solution will not change the target distribution of the Metropolis-Hastings scheme.

Our proposed algorithm for taking advantage of the CLE approximation, which we call delayed acceptance PMMH (daPMMH), is outlined in Algorithm 1; the algorithm which takes advantage of the LNA is a slight simplification of this. Both algorithms have the following basic structure.

First a candidate set of parameter values is proposed, then a decision is made whether to accept or reject these values based on a MH step with target density $p_a(c|\mathbf{y}) \propto p_a(\mathbf{y}|c)p(c)$, which is the posterior density of parameters under the approximate model (for example, the LNA or the CLE); here $p_a(\mathbf{y}|c)$ represents the marginal likelihood under the approximate model. If the proposed parameter values are accepted at this first stage then they undergo another MH step with target density $p(c|\mathbf{y}) \propto p(\mathbf{y}|c)p(c)$, which is the marginal posterior density under the MJP. The idea here is that the first stage weeds out 'poor' parameter values. Consequently, the computationally expensive SMC algorithm for the MJP is only implemented for 'good' parameter values which are likely to be accepted at the second stage.

When the CLE is used as the approximate model the marginal likelihood $p_a(\mathbf{y}|c)$ is not available analytically, so we replace it with an unbiased estimate $\widehat{p}_a(\mathbf{y}|c)$ obtained from an SMC scheme which targets $p_a(\mathbf{x}|\mathbf{y}, c)$, the conditional density of the latent states under the approximate model, given the observed data and the parameter values. We therefore have to run a particle filter at both stages of the daPMMH algorithm, as one is always needed at stage 2 to give an unbiased estimate $\widehat{p}(\mathbf{y}|c)$ of the MJP marginal likelihood $p(\mathbf{y}|c)$. We note, however, that despite the CLE requiring a run of an SMC scheme to obtain $\widehat{p}_a(\mathbf{y}|c)$ this may still be much faster to run than the SMC scheme for the MJP (with the same number of particles).

Our daPMMH algorithm is an extension of the delayed acceptance MH (daMH) algorithm of Christen and Fox (2005), which is a version of the 'surrogate transition method' of Liu (2001). Specifically, we have extended the daMH algorithm by replacing all intractable marginal likelihoods by unbiased estimates obtained from appropriate SMC schemes. Our extension of the daMH algorithm to an intractable likelihood at Stage 1 is essential when the approximate model is the CLE since the marginal likelihood under the CLE is intractable. However, when the LNA is chosen as the approximate model this extra level of complexity is not necessary; we simply replace the marginal likelihood estimates $\widehat{p}_a(\mathbf{y}|c)$ in Algorithm 1 with the exact values $p_a(\mathbf{y}|c)$ since these are available numerically (see the section " Marginal likelihood under the linear noise approximation" of Appendix for details). Despite replacing the intractable marginal likelihoods by unbiased estimates, our daPMMH algorithm still targets the (exact) posterior density of the parameters under the MJP, $p(c|\mathbf{y})$, as we outline in Sect. 3.3.1. Note that in an independent technical report, Smith (2011) proved that the daPMMH algorithm has $p(c|\mathbf{y})$ as its target density when the marginal likelihood under the approximate model is tractable. In Sect. 3.3.1 we generalise the argument of Smith (2011) to the case of an SMC-based marginal likelihood estimate for the approximate model.

3.3.1 Validity of delayed acceptance PMMH

In this section we show that the daPMMH algorithm (Algorithm 1) is a valid MCMC scheme which targets a distribution that admits $p(c|\mathbf{y})$ as a marginal distribution.

We first define some notation and an extended state-space. Let $F : \mathbb{R}^2 \to [0, 1]$ be any function satisfying the following.

$$aF[a, a^*] = a^*F[a^*, a] \tag{16}$$

$$F[ba, ba^*] = F[a, a^*]. \tag{17}$$

An example of F is the Metropolis-Hastings acceptance probability $F[a, a^*] = \min(1, a^*/a)$, with $a = p(c|\mathbf{y})q(c^*|c)$ and $a^* = p(c^*|\mathbf{y})q(c|c^*)$. More generally, F defines an acceptance probability that admits a chain with invariant density a, a *joint density* (known up to an arbitrary constant) *on the current value in the chain and the next proposal*. Condition (16) ensures that detailed balance is satisfied with respect to a, and Condition (17) ensures that the target density need only be known up to a fixed constant.

Let U be a vector of auxiliary random variables, sampled conditional on the parameters according to $q_a(u|c)$, and let $\hat{p}(c, u)$ and $\hat{p}_a(c, u)$ be two approximations to the posterior which depend on U, with $\hat{p}$ unbiased up to a fixed constant, $k > 0$:

$$\int \hat{p}(c, u)\, q_a(u|c)\, du = k\; p(c|\mathbf{y}). \tag{18}$$

Note that for notational simplicity, we have dropped dependence of $\hat{p}(c, u)$ and $\hat{p}_a(c, u)$ on the data $\mathbf{y}$. For further clarity of exposition we adopt the shorthand

$$\hat{p} := \hat{p}(c, u), \quad \hat{p}^* := \hat{p}(c^*, u^*), \; \hat{p}_a := \hat{p}_a(c, u),$$
$$\hat{p}_a^* := \hat{p}_a(c^*, u^*), \quad q_a := q_a(u|c), \; q_a^* = q_a(u^*|c^*).$$

Our delayed-acceptance Markov chain proposes according to $q(c^*|c)q_a^*$ and accepts with a probability of

$$\alpha\left(c, u; c^*, u^*\right) = F\left[\hat{p}_a\, q(c^*|c), \hat{p}_a^* q(c|c^*)\right] \times F\left[\frac{\hat{p}}{\hat{p}_a}, \frac{\hat{p}^*}{\hat{p}_a^*}\right].$$

Our chain targets the joint posterior $\hat{p}(c, u)q_a(u|c)$ so that, by (18), the marginal distribution for c is the posterior $p(c|\mathbf{y})$. To show that $\hat{p}(c, u)q_a(u|c)$ is indeed the invariant distribution of the chain it is sufficient to show that our chain satisfies detailed balance with respect to this posterior. Since rejection moves ($c^* \leftarrow c$, $u^* \leftarrow u$) automatically satisfy detailed balance we need only consider moves where the proposal is accepted. Now

$$\hat{p}\; q_a\; q(c^*|c)\; q_a^* = \hat{p}_a\; q(c^*|c) \times \frac{\hat{p}\; q_a\; q_a^*}{\hat{p}_a}.$$

By (16),

$$\hat{p}_a\; q(c^*|c)\; F\left[\hat{p}_a\; q(c^*|c), \hat{p}_a^*\; q(c|c^*)\right]$$
$$= \hat{p}_a^*\; q(c|c^*)\; F\left[\hat{p}_a^*\; q(c|c^*), \hat{p}_a\; q(c^*|c)\right].$$

Also, by (17) then (16) then (17) again,

$$\begin{aligned}
&\frac{\hat{p}\; q_a\; q_a^*}{\hat{p}_a} \times F\left[\frac{\hat{p}}{\hat{p}_a}, \frac{\hat{p}^*}{\hat{p}_a^*}\right] \\
&\quad = \frac{\hat{p}\; q_a\; q_a^*}{\hat{p}_a} \times F\left[\frac{\hat{p}\; q_a\; q_a^*}{\hat{p}_a}, \frac{\hat{p}^*\; q_a\; q_a^*}{\hat{p}_a^*}\right] \\
&\quad = \frac{\hat{p}^*\; q_a\; q_a^*}{\hat{p}_a^*} \times F\left[\frac{\hat{p}^*\; q_a\; q_a^*}{\hat{p}_a^*}, \frac{\hat{p}\; q_a\; q_a^*}{\hat{p}_a}\right] \\
&\quad = \frac{\hat{p}^*\; q_a\; q_a^*}{\hat{p}_a^*} \times F\left[\frac{\hat{p}^*}{\hat{p}_a^*}, \frac{\hat{p}}{\hat{p}_a}\right].
\end{aligned}$$

Thus

$$\hat{p}\; q_a\; q(c^*|c)\; q_a^*\; \alpha(c, u; c^*, u^*)$$
$$= \hat{p}^*\; q_a^*\; q(c|c^*)\; q_a\; \alpha(c^*, u^*; c, u),$$

as required. When our Stage 1 approximation is deterministic (using the LNA) then it is independent of U. Otherwise, when we use the CLE at Stage 1, our two estimates of the posterior are independent, i.e. U is split into two independent vectors, U_1 and U_2, with $\hat{p}_a$ a function of U_1 only and $\hat{p}$ a function of U_2 only. However, for the algorithm to work we only need to be able to simulate U_1 (for Stage 1) and then, if required, $U_2|U_1 = u_1$ (for Stage 2); the independence is not necessary. Indeed a higher Stage 2 acceptance rate might be obtainable if it were possible to make $\hat{p}_a(c, U)$ and $\hat{p}(c, U)$ positively correlated. Unfortunately we cannot see any obvious method for constructing correlated estimators based upon the CLE and the MJP.

3.3.2 Comments on efficiency

Christen and Fox (2005) note that with a fast approximate model daMH algorithms are less computationally expensive—that is, they exhibit lower CPU times for the same number of iterations—than standard MH algorithms that do not employ delayed acceptance. They also note that daMH algorithms are less statistically efficient than standard MH algorithms that do not employ delayed acceptance. Here statistical efficiency relates to the mixing of the Markov chain, and can be measured by the effective sample size (ESS), the number of independent samples that are equivalent in information content to the actual number of dependent samples from the Markov chain. Clearly, computational time is dictated by the speed with which $p_a(\mathbf{y}|c)$ (or its estimate $\widehat{p}_a(\mathbf{y}|c)$) is computed, and statistical efficiency is dictated by the accuracy of the approximation $p_a(\mathbf{y}|c)$ or $\widehat{p}_a(\mathbf{y}|c)$ to $p(\mathbf{y}|c)$. For example, $p_a(\mathbf{y}|c)$ under the LNA will be faster to compute than $\widehat{p}_a(\mathbf{y}|c)$ under the CLE since the latter requires a run of an SMC algorithm. However, we might expect the CLE (at least with a small Euler time-step) to provide a better approximation to the MJP than the LNA, since the LNA is, in

some sense, a simplified version of the CLE. Increasing the time-step Δt in the CLE will decrease the computation time but should also decrease the accuracy of the approximation; the trade-off in terms of computational efficiency between these two factors merits further investigation.

Another factor which will affect statistical efficiency is the variability associated with the SMC-based estimate of marginal likelihood $\widehat{p}_a(\mathbf{y}|c)$. An algorithm using $\widehat{p}_a(\mathbf{y}|c)$ will be less statistically efficient than an idealised algorithm which uses $p_a(\mathbf{y}|c)$ (for the same approximate model). We might expect, therefore, that using the LNA as the approximate model, with its tractable marginal likelihood, may lead to increased statistical efficiency over the CLE-based approximation, although this depends on the accuracy of the LNA.

The daPMMH scheme (using either the LNA or CLE) requires specification of a number of particles N to be used in the SMC scheme at Stage 2. As noted by Andrieu and Roberts (2009), the mixing efficiency of the PMMH scheme decreases as the variance of the estimated marginal likelihood increases. This problem can be alleviated at the expense of greater computational cost by increasing N. This therefore suggests an optimal value of N and finding this choice is the subject of Pitt et al. (2012) and Doucet et al. (2013). The latter suggest that N should be chosen so that the variance in the noise in the estimated log-posterior is around 1. Pitt et al. (2012) note that the penalty is small for a value between 0.25 and 2.25. We therefore recommend performing an initial pilot run of daPMMH to obtain an estimate of the posterior mean for the parameters c, denoted $\hat{c}$. The value of N should then be chosen so that $\mathrm{Var}(\log p(\mathbf{y}|\hat{c}))$ is around 1–1.5. When the CLE is used as a surrogate model, we must also specify a number of particles (say N_1) to be used in Stage 1. For simplicity, we take $N_1 = N$. Provided the CLE is a reasonable approximation to the MJP, we may expect that N_1 provides a suitable trade-off between computational cost and accuracy (in terms of the variance of the estimated marginal likelihood under the CLE).

In the next section we show empirically that our daPMMH algorithm (with either the CLE or the LNA as the approximate model) can lead to improvements in overall computational efficiency (in terms of ESS normalised by CPU time) over a vanilla PMMH scheme for the MJP.

4 Applications

4.1 Lotka–Volterra

Following Boys et al. (2008), we consider first a Lotka–Volterra model of predator and prey interaction comprising three reactions:

$$\mathcal{R}_1: \quad \mathcal{X}_1 \xrightarrow{c_1} 2\mathcal{X}_1$$

$$\mathcal{R}_2: \quad \mathcal{X}_1 + \mathcal{X}_2 \xrightarrow{c_2} 2\mathcal{X}_2$$

$$\mathcal{R}_3: \quad \mathcal{X}_2 \xrightarrow{c_3} \emptyset.$$

For simplicity of notation we drop the explicit dependence of the current state $X = (X_1, X_2)'$ and the deterministic approximation $z = (z_1, z_2)'$ on time, t. The stoichiometry matrix is given by

$$S = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

and the associated hazard function is

$$h(X, c) = (c_1 X_1, c_2 X_1 X_2, c_3 X_2)'.$$

The diffusion approximation can be calculated by substituting S and $h(X, c)$ into the CLE (1) to give respective drift and diffusion coefficients of

$$\alpha(X, c) = \begin{pmatrix} c_1 X_1 - c_2 X_1 X_2 \\ c_2 X_1 X_2 - c_3 X_2 \end{pmatrix},$$

$$\beta(X, c) = \begin{pmatrix} c_1 X_1 + c_2 X_1 X_2 & -c_2 X_1 X_2 \\ -c_2 X_1 X_2 & c_2 X_1 X_2 + c_3 X_2 \end{pmatrix}.$$

For the linear noise approximation, the Jacobian matrix F_t is given by

$$F_t = \begin{pmatrix} c_1 - c_2 z_2 & -c_2 z_1 \\ c_2 z_2 & c_2 z_1 - c_3 \end{pmatrix}.$$

We simulated a synthetic dataset by generating 50 observations at integer times using the Gillespie algorithm with initial conditions $x_1 = (70, 80)'$ and parameter values $c = (1.0, 0.005, 0.6)'$ taken from Wilkinson (2012). Predator values were discarded leaving 50 observations on prey only. These were then corrupted via an error distribution for which the marginal likelihood under the LNA is intractable:

$$Y_t \sim \mathrm{Poisson}(x_{1,t}), \qquad t = 1, 2, \ldots, 50.$$

A tractable approximation to the true marginal likelihood under the MJP, for use in Stage 1 of the delayed acceptance scheme was obtained using the LNA as described in Sect. 3.2. In what follows, for simplicity, we assume that the latent initial state x_1 is known.

For brevity, we refer to the MCMC algorithm targeting the posterior under the MJP that uses the LNA inside the delayed acceptance PMMH scheme as *daPMMH-LNA*. Similarly, when using the CLE inside the delayed acceptance scheme we refer to this as *daPMMH-CLE*. Finally, we designate the vanilla PMMH scheme without delayed rejection as *PMMH*. Using independent Uniform $U(-8, 8)$ priors for each $\log(c_i)$ we performed a pilot run of the PMMH scheme with 50 particles to give an approximate covariance matrix $\widehat{\mathrm{Var}}(c)$ and approximate posterior mean $\hat{c}$. Further pilot runs were then implemented with c fixed at $\hat{c}$ and numbers of particles ranging from 50 to 250. We found that using 200 particles

gave the variance in the noise in the estimated log-posterior as 1.16. We therefore took $N = 200$ particles for the main monitoring runs, which consisted of 2×10^5 iterations of each scheme, with the $\log(c_i)$ updated in a single block using a Gaussian random walk proposal kernel. For PMMH, we followed the practical advice of Sherlock et al. (2013) and used an innovation variance matrix given by $\lambda \frac{2.38^2}{3} \widehat{\mathrm{Var}}(c)$ with λ tuned to give an acceptance rate of around 10 %. We tried a range of λ values and report results for $\lambda = 0.7$ which gave an acceptance rate of 9.4 %. For daPMMH-CLE and daPMMH-LNA, we found that using $\lambda = 1$ and $\lambda = 3$ (respectively) gave an improved overall efficiency (compared with simply using $\lambda = 0.7$). Intuitively, as computation of an estimate of marginal likelihood under the CLE and an approximation to the marginal likelihood under the LNA is extremely cheap relative to the MJP, larger moves should be tried at Stage 1. For daPMMH-CLE, we considered three levels of discretisation, namely, $\Delta t = 0.2, 0.125, 0.0625$. The cost of computing either an estimate of marginal likelihood (under the CLE) or an approximation to the marginal likelihood (under the LNA) scales roughly as $1 : 20 : 30 : 58 : 362$ for LNA : CLE($\Delta t = 0.2$) : CLE($\Delta t = 0.125$) : CLE($\Delta t = 0.0625$) : MJP. All algorithms are coded in C and were run on a desktop computer with a 2.83 GHz clock speed.

Figure 1 summarises the output of the PMMH scheme (consistent with the output of the delayed acceptance schemes, not reported). We also give kernel density estimates of the marginal parameter posteriors under the LNA and CLE (for each discretisation choice). That is, we ran daPMMH-LNA and daPMMH-CLE without performing the Stage 2 correction. When working with the CLE, smaller Euler time steps appear to give a better approximation. The effect of this choice on overall efficiency can be seen in Table 1. Here, we report Stage 1 acceptance rate α_1, Stage 2 acceptance rate $\alpha_{2|1}$, the CPU time, the minimum (over the 3 parameters) effective sample size (ESS_{min}) and minimum effective sample size per second, relative to the corresponding value obtained from the vanilla PMMH scheme. Whilst the daPMMH-CLE scheme gives an improvement in overall efficiency (as measured by relative ESS_{min} per second) for all values of Δt employed, the effect of the discretisation is clear. The marginal likelihood under the CLE approaches that under the MJP as Δt decreases, resulting in greater statistical efficiency of the daPMMH-CLE scheme. This can also be seen by inspecting the Stage 2 acceptance probability reported in Table 1. Naturally, this improvement comes at a greater computational cost suggesting an optimal value of Δt between 0.2 and 0.0625 for this example. Perhaps counter-intuitively, the CPU time for $\Delta t = 0.2$ is actually greater than that for $\Delta t = 0.125$. Whilst all three approximate posteriors that are derived from the CLE are wider than that derived from the MJP, the approximate posterior with $\Delta t = 0.2$ is by far the widest. Consequently the Stage 1 acceptance rate is much higher and the computationally intensive Stage 2 calculation is performed more often. Further insight into this result can be gained from Fig. 2, which plots estimates of the marginal likelihood (on the log-scale) under PMMH against the corresponding value obtained under each approximation, for 10,000 values of c sampled from the posterior $p(c|\mathbf{y})$. The Stage 1 and 2 acceptance rates depend only on the estimates of the log-likelihood at the proposed and current values through their difference. Thus the efficiency of the algorithm is unaffected by any fixed shift of the points from the line through the origin with a slope of one. However, variability about a line with this slope is important and we see greater variability in the estimates obtained for $\Delta t = 0.2$ resulting in a reduction in statistical efficiency for the daPMMH-CLE ($\Delta t = 0.2$) scheme, with proposed values that were accepted at Stage 1 being rejected at Stage 2.

The daPMMH-LNA scheme on the other hand requires minimal tuning. The LNA gives an analytic form for the (approximate) marginal likelihood and therefore does not

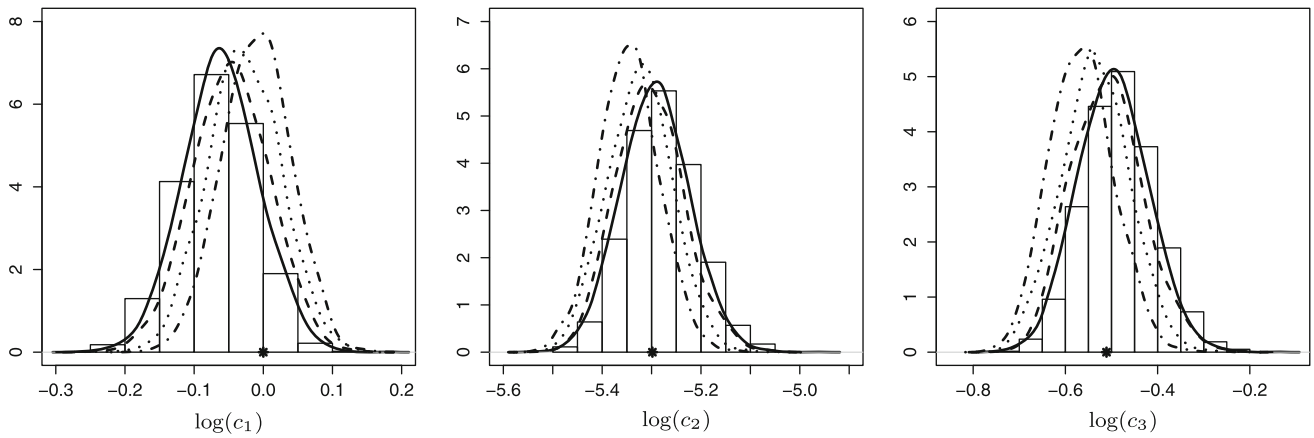


Fig. 1 Lotka–Volterra model. Marginal posterior distributions under the MJP (histogram), LNA (*solid line*) and CLE with $\Delta t = 0.0625$ (*dashed line*), $\Delta t = 0.125$ (*dotted line*), $\Delta t = 0.2$ (*dot-dashed line*). True values of each $\log(c_i)$ are indicated *asterisk*

Table 1 Lotka–Volterra model

Algorithm	α_1	$\alpha_{2\|1}$	CPU time (s)	ESS_{min}	Rel. ESS_{min}/s
PMMH	0.094	1.000	74,850	2,186	1.00
daPMMH-CLE ($\Delta t = 0.2$)	0.123	0.142	15,167	485	1.10
daPMMH-CLE ($\Delta t = 0.125$)	0.105	0.278	14,814	867	2.00
daPMMH-CLE ($\Delta t = 0.0625$)	0.109	0.327	21,230	948	1.53
daPMMH-LNA	0.031	0.464	2,581	835	11.08

Stage 1 acceptance rate α_1, Stage 2 acceptance rate $\alpha_{2|1}$, CPU time (to the nearest second), minimum effective sample size (ESS_{min}, to the nearest whole number) and minimum effective sample size per second, relative to the corresponding value obtained from the vanilla PMMH scheme. All values are based on 10^5 iterations

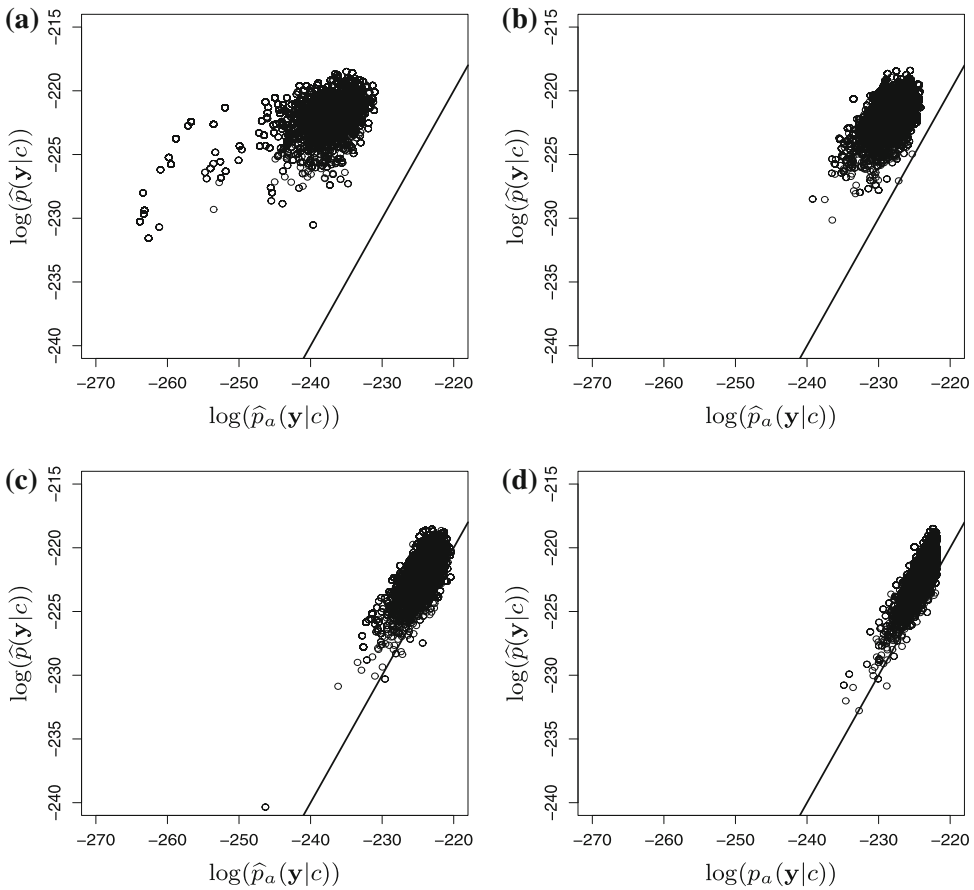


Fig. 2 Log-marginal likelihood estimates under the MJP ($\log(\widehat{p}(\mathbf{y}|c))$) against the corresponding log-marginal likelihood estimate under (**a**) the CLE ($\Delta t = 0.2$), (**b**) the CLE ($\Delta t = 0.125$), (**c**) the CLE ($\Delta t = 0.0625$) and (**d**) the LNA. All plots are obtained using 10,000 values of c sampled from the posterior $p(c|\mathbf{y})$ for the Lotka–Volterra model

require implementation of a particle filter during the first Stage of the delayed acceptance scheme. Moreover, the LNA solution involves solving a set of ODEs, for which standard routines, such as the `lsoda` package (Petzold 1983), exist. Therefore, pre-specification of a suitable time discretisation is not required. We find for this example that the daPMMH-LNA scheme outperforms the vanilla PMMH scheme by a factor of more than 10. In what follows, we focus on the daPMMH-LNA scheme.

4.2 Gene expression

Here, we consider a simple model of gene expression involving three biochemical species (DNA, mRNA, protein) and four reaction channels (transcription, mRNA degradation, translation, protein degradation):

$$\mathcal{R}_1: \quad DNA \xrightarrow{\kappa_{R,t}} DNA + R$$
$$\mathcal{R}_2: \quad R \xrightarrow{\gamma_R} \emptyset$$
$$\mathcal{R}_3: \quad R \xrightarrow{\kappa_P} R + P$$
$$\mathcal{R}_4: \quad P \xrightarrow{\gamma_P} \emptyset.$$

This system has been analysed by Komorowski et al. (2009) among others, and we therefore adopt the same notation to aid the exposition.

Let $X_t = (R_t, P_t)'$ denote the system state at time t, where R_t and P_t are the respective number of mRNA and protein molecules. As in Komorowski et al. (2009), we take $\kappa_{R,t}$ to be the time dependent transcription rate of the gene. Specifically,

$$\kappa_{R,t} = b_0 \exp\left(-b_1(t-b_2)^2\right) + b_3$$

so that transcription rate increases for $t < b_2$ and tends to the baseline b_3 for $t > b_2$. We denote the vector of unknown parameters by

$$c = (\gamma_R, \gamma_P, \kappa_P, b_0, b_1, b_2, b_3)'$$

and our goal is to perform inference for these parameters. The stoichiometry matrix associated with the system is given by

$$S = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

and the associated hazard function is

$$h(X_t, c) = (\kappa_{R,t}, \gamma_R R_t, \kappa_P R_t, \gamma_P P_t)'.$$

For the linear noise approximation, we have the Jacobian matrix as

$$F_t = \begin{pmatrix} -\gamma_R & 0 \\ \kappa_P & -\gamma_P \end{pmatrix}.$$

We simulated a synthetic dataset by generating observations every 15 min for 25 h (giving 100 observations in total) noting that care must be taken when simulating from the MJP representation of this system, due to the time dependent hazard of reaction $\mathcal{R}_1$. We used initial conditions of $x_1 = (10, 150)'$ and parameter values $c = (0.44, 0.52, 10, 15, 0.4, 7, 3)'$ with units of time in hours. As in Komorowski et al. (2009) we created a challenging data-poor scenario by discarding observations on mRNA levels and corrupting the remaining protein observations with additive Gaussian noise:

$$Y_t \sim \mathrm{N}(P_t, \sigma^2), \qquad t = 1, 2, \ldots, 100.$$

We took $\sigma = 10$ and assume that this quantity is unknown. We therefore augment the parameter vector c to include σ. The data are shown in Fig. 3.

For each rate constant, we assumed the same prior distributions as in Komorowski et al. (2009) including informative priors for the degradation rates to ensure identifiability. Specifically, we have that

$$\begin{aligned} &\gamma_R \sim \Gamma(19.36, 44), && \gamma_P \sim \Gamma(27.04, 52), \\ &\kappa_P \sim \mathrm{Exp}(0.01), && b_0 \sim \mathrm{Exp}(0.01), \\ &b_1 \sim \mathrm{Exp}(1), && b_2 \sim \mathrm{Exp}(0.1), \\ &b_3 \sim \mathrm{Exp}(0.01), && \sigma \sim \mathrm{Exp}(0.01) \end{aligned}$$

where $\Gamma(a, b)$ denotes the Gamma distribution with mean a/b and $\mathrm{Exp}(b)$ denotes the Exponential distribution with mean $1/b$). For simplicity, we fixed the initial latent states at their true values. We performed a pilot run of the PMMH scheme with 50 particles to give an approximate covariance matrix $\widehat{\mathrm{Var}}(c)$ and approximate posterior mean $\hat{c}$. By performing further pilot runs we found that using 250 particles gave the variance in the noise in the estimated log-posterior as 1.54. We therefore took $N = 250$ particles for the main monitoring runs, which typically consisted of 2×10^5 iterations of each scheme, with the $\log(c_i)$ updated in a single block using a Gaussian random walk proposal kernel. We used an innovation variance matrix given by $\lambda \frac{2.38^2}{3} \widehat{\mathrm{Var}}(c)$. For PMMH, further pilot runs were performed to determine an appropriate scaling λ. We used $\lambda = 0.6$ (which gave an acceptance rate of around 8 %) for the main run. The cost of computing an approximation to the marginal likelihood (under the LNA) versus computing an estimate of marginal likelihood under the MJP scales roughly as 1:780 for LNA: MJP and we might therefore expect that a larger value of λ will be optimal for daPMMH-LNA. In order to investigate effect of λ on the daPMMH-LNA scheme, we report results for $\lambda = 0.6, 1, 2, 3, 4$.

Figure 4 summarises the output of the PMMH scheme which we find to be consistent with the output of the daPMMH-LNA scheme (not reported). We also give ker-

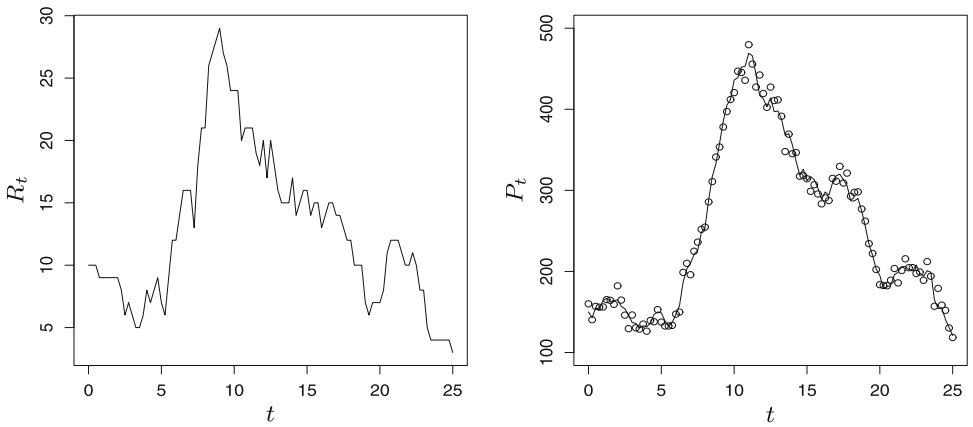


Fig. 3 A single realisation of the gene expression system obtained using the first reaction method. Protein numbers used in the artificial dataset are shown as *circles*

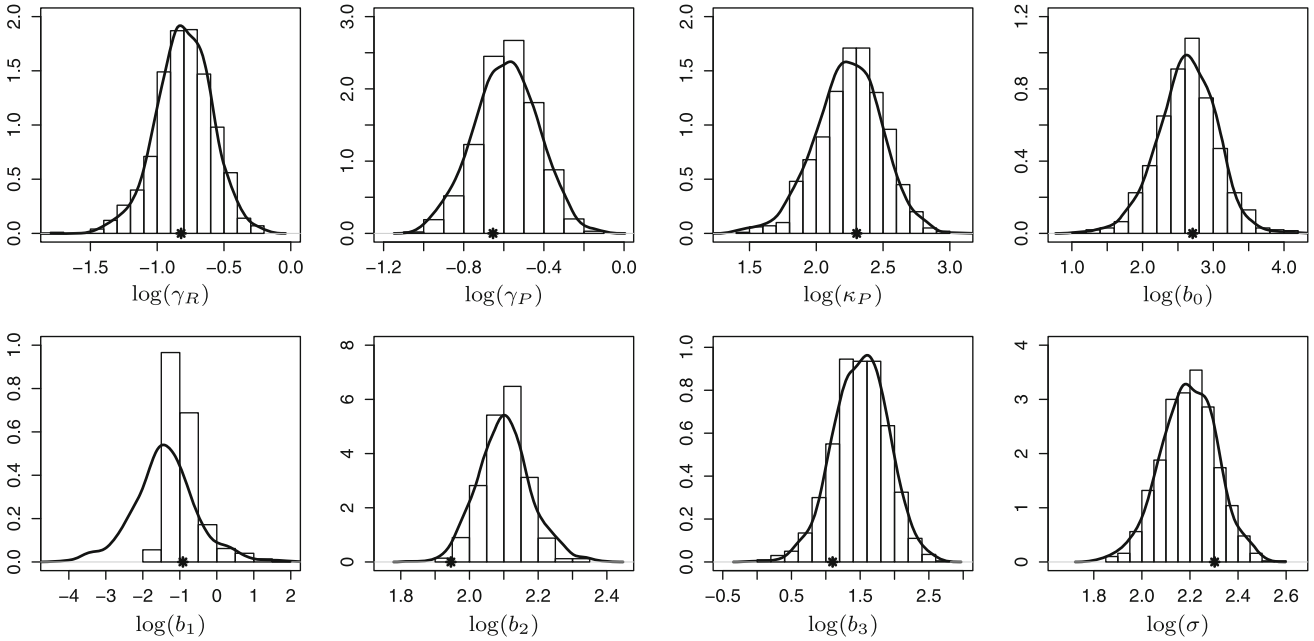


Fig. 4 Gene expression model. Marginal posterior distributions under the MJP (histograms) and LNA (*solid line*). True values of each $\log(c_i)$ are indicated *asterisk*

Table 2 Gene expression model

Algorithm	α_1	$\alpha_{2\|1}$	CPU time (s)	ESS_{min}	Rel. ESS_{min}/s
PMMH	0.077	1.000	350,657	524	1.00
daPMMH-LNA ($\lambda = 0.6$)	0.218	0.198	77,704	316	2.72
daPMMH-LNA ($\lambda = 1$)	0.137	0.178	50,840	394	5.19
daPMMH-LNA ($\lambda = 2$)	0.051	0.163	20,155	246	8.18
daPMMH-LNA ($\lambda = 3$)	0.029	0.149	11,667	153	8.76
daPMMH-LNA ($\lambda = 4$)	0.023	0.182	9,518	120	8.44

Stage 1 acceptance rate α_1, Stage 2 acceptance rate $\alpha_{2|1}$, CPU time (to the nearest second), minimum effective sample size (ESS_{min}, to the nearest whole number) and minimum effective sample size per second, relative to the corresponding value obtained from the vanilla PMMH scheme. All values are based on 2×10^5 iterations

nel density estimates of the marginal parameter posteriors under the LNA. The posterior samples appear to be consistent with the true values that produced the data although we see some discrepancy between the LNA and MJP posteriors. Table 2 shows Stage 1 acceptance rate α_1, Stage 2 acceptance rate $\alpha_{2|1}$, CPU time, minimum (over the parameters) effective sample size (ESS_{min}) and minimum effective sample size per second, relative to the corresponding value obtained from the PMMH scheme. The effect of increasing the scaling parameter λ (which in turn increases the innovation variance for the Gaussian random walk update) can clearly be seen. When $\lambda = 3$ we see an 8 fold improvement in overall efficiency (as measured by relative ESS_{min} per second). The result is relatively robust to the choice of λ, with a relative ESS_{min} per second of 2.72 when using the same scaling as PMMH ($\lambda = 0.6$).

The accuracy of the LNA can be assessed through inspection of Figs. 4 and 5. There is a noticeable discrepancy in the marginal posteriors for $\log(b_1)$ and $\log(b_2)$. Despite this, Fig. 5 suggests that the LNA provides a reasonable approximation to the MJP in regions of high posterior density, and we recorded an empirical Stage 2 acceptance probability of around 0.18.

4.3 Epidemic model

Finally, we consider a Susceptible–Infected–Removed (SIR) epidemic model involving two species (susceptibles $\mathcal{X}_1$ and infectives $\mathcal{X}_2$) and two reaction channels (infection of a susceptible and removal of an infective):

$$\mathcal{R}_1 : \quad \mathcal{X}_1 + \mathcal{X}_2 \xrightarrow{\beta} 2\mathcal{X}_2$$
$$\mathcal{R}_2 : \quad \mathcal{X}_2 \xrightarrow{\gamma} \emptyset.$$

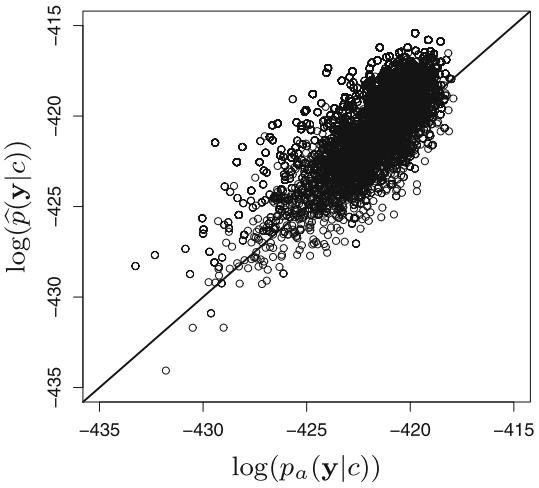


Fig. 5 Log-marginal likelihood estimates $\log(\widehat{p}(\mathbf{y}|c))$ under the MJP against the corresponding log-marginal likelihood estimate under the LNA, using 10,000 values of c sampled from the posterior $p(c|\mathbf{y})$ for the gene expression model

The system can be seen as a special case of the Lotka–Volterra system with $c_1 = 0$. We let $c = (\beta, \gamma)'$ denote the unknown parameter vector. The stoichiometry matrix is given by

$$S = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix}$$

and the associated hazard function is

$$h(X, c) = (\beta X_1 X_2, \gamma X_2)'$$

where $X = (X_1, X_2)'$ denotes the state of the system at time t. For the linear noise approximation, the Jacobian matrix F_t is given by

$$F_t = \begin{pmatrix} -\beta z_2 & -\beta z_1 \\ \beta z_2 & \beta z_1 - \gamma \end{pmatrix}$$

where $z = (z_1, z_2)'$ is the state at time t of the deterministic process satisfying (6).

We consider the Abakaliki small pox dataset given in Bailey (1975) and studied by O'Neill and Roberts (1999), Fearnhead and Meligkotsidou (2004) and Boys and Giles (2007) among others. Page 125 of Bailey (1975) provides a complete set of 29 inter-removal times, measured in days, from a smallpox outbreak in a community of 120 individuals in Nigeria. We report the data here as the days on which the removal of individuals actually took place, with the first day set to be time 0 (Table 3). We assume an SIR model for the data with observations being equivalent to daily measurements of $X_1 + X_2$ (as there is a fixed population size). In addition, and for simplicity, we assume that a single individual remained infective just after the first removal occurred. We analyse the data under the assumption of no measurement error. This assumption can be incorporated into the PMMH algorithm by calculating the un-normalised weight in step 2(b) of the SMC scheme as

$$w_{t+1}^{*i} = \begin{cases} 1, & x_{t+1}^i = y_{t+1} \\ 0, & \text{otherwise} \end{cases}$$

The marginal likelihood under the LNA can be computed using the algorithm described in the section " Marginal likelihood under the linear noise approximation" of Appendix with $G' = (1, 1)$ and $\Sigma = 0$. Note that for this example, the cost of computing the LNA marginal likelihood versus an estimate of marginal likelihood under the MJP scales roughly as 1:34 for LNA : MJP.

We followed Fearnhead and Meligkotsidou (2004) by taking $\beta \sim \Gamma(10, 10^4)$ and $\gamma \sim \Gamma(10, 10^2)$ a priori. A pilot run of the PMMH scheme with 500 particles was used to give an approximate covariance matrix $\widehat{\text{Var}}(c)$ and approximate posterior mean $\hat{c}$. By performing further pilot runs we found that using 2000 particles gave the variance in the noise in the estimated log-posterior as 1.25. We therefore took $N = 2000$ particles for the main monitoring runs, which typically consisted of 10^5 iterations of each scheme, with the $\log(c_i)$ updated in a single block using a Gaussian random walk proposal kernel with innovation variance $\lambda \frac{2.38^2}{3} \widehat{\text{Var}}(c)$. For PMMH, a number of short pilot runs suggested that $\lambda = 1.1$ (which gave an acceptance rate of 0.23) was close to optimal.

Figure 6 gives marginal posterior densities under the MJP (using the output of the PMMH scheme) and the LNA (using the output of the daPMMH-LNA scheme without Stage 2). We see that the LNA substantially underestimates the uncertainty in β. Use of the LNA as a surrogate model in this case will likely lead to rejected parameter draws at Stage 1 that would otherwise be accepted at Stage 2. We alleviate this problem by scaling the log marginal likelihood under the LNA by an amount $1/\tau$, where τ is chosen to maximise the efficiency of the delayed acceptance scheme. Specifically, we replace $p_a(\mathbf{y}|c)$ in Algorithm 1 with $p_a(\mathbf{y}|c)^{\frac{1}{\tau}}$. To determine

Table 3 Abakaliki smallpox data

Day	0	13	20	22	25	26	30	35	38	40	42	47
No. of removals	1	1	1	1	3	1	1	1	1	2	2	1
Day	50	51	55	56	57	58	60	61	66	71	76	
No. of removals	1	1	2	1	1	1	2	1	2	1	1	

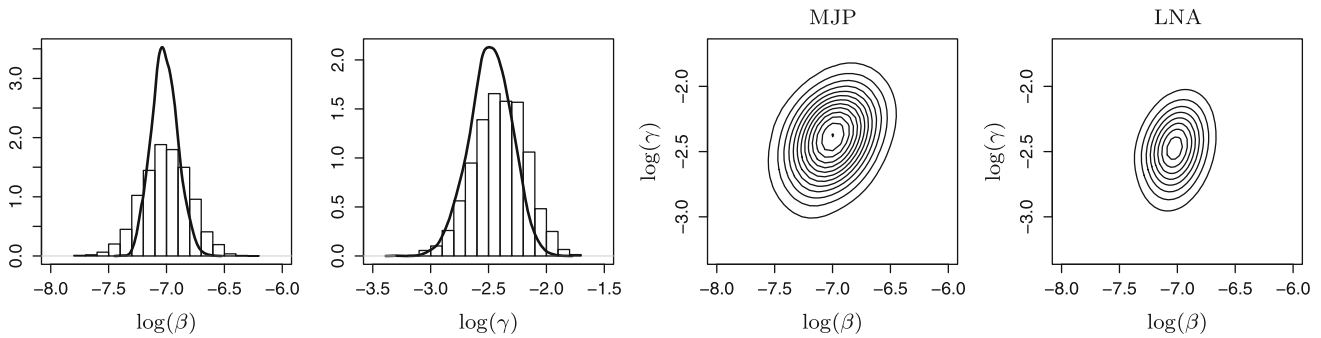


Fig. 6 Epidemic model. Marginal posterior distributions under the MJP (histograms) and LNA (*solid line*), and contour plots of the joint posterior under the MJP (*left*) and LNA (*right*)

Table 4 Epidemic model

Algorithm	α_1	$\alpha_{2\|1}$	CPU time (s)	ESS_{min}	Rel. ESS_{min}/s
PMMH	0.226	1.000	4,981	7,469	1.00
daPMMH-LNA ($\tau = 1$, $\lambda = 1.1$)	0.252	0.402	1,208	1,478	0.82
daPMMH-LNA ($\tau = 4$, $\lambda = 1.1$)	0.345	0.509	1,634	3,444	1.41
daPMMH-LNA ($\tau = 5$, $\lambda = 1.1$)	0.358	0.499	1,698	4,374	1.72
daPMMH-LNA ($\tau = 6$, $\lambda = 1.1$)	0.372	0.488	1,763	2,831	1.07
daPMMH-LNA ($\tau = 5$, $\lambda = 3$)	0.180	0.476	890	2920	2.19
daPMMH-LNA ($\tau = 5$, $\lambda = 4$)	0.144	0.471	762	2471	2.16
daPMMH-LNA ($\tau = 5$, $\lambda = 5$)	0.120	0.468	649	2008	2.06

Stage 1 acceptance rate α_1, Stage 2 acceptance rate $\alpha_{2|1}$, CPU time (to the nearest second), minimum effective sample size (ESS_{min}, to the nearest whole number) and minimum effective sample size per second, relative to the corresponding value obtained from the vanilla PMMH scheme. All values are based on 10^5 iterations

an appropriate value for τ, we fixed the scaling λ at 1.1 and ran the daPMMH-LNA scheme for τ in the range $[1, 10]$. Table 4 reports results for $\tau \in \{1, 4, 5, 6\}$. We see that as τ increases so does the Stage 1 acceptance rate, resulting in an increase in CPU time (as the expensive MJP simulator is run more often). However, the Stage 2 acceptance rate also increases, suggesting an optimal value of τ. We found that $\tau = 5$ is optimal for the range considered. We therefore fixed $\tau = 5$ and varied the scaling λ. For $\lambda \in \{3, 4, 5\}$ it is possible to achieve a 2-fold increase in efficiency over PMMH.

5 Discussion

We have proposed two delayed acceptance Particle Mar-ginal Metropolis-Hastings algorithms, analogues of the delayed acceptance Metropolis-Hastings scheme of Christen and Fox (2005). We have shown that both lead to a chain with the desired stationary distribution and applied them to the problem of parameter estimation in Markov jump processes with state-dependent rate parameters. In both analogues the true posterior that is used in Christen and Fox (2005) is replaced with an unbiased approximation obtained through a particle filter. In the second analogue the fast deterministic approximation is replaced with a relatively fast stochastic approximation that is also obtained via a particle filter. The need for such an approach is motivated by the potentially huge computational cost of performing particle MCMC for the MJP directly, where each iteration requires implementation of a particle filter with N particles, and a complete run of the stochastic simulation algorithm is required for each particle.

The delayed acceptance PMMH scheme aims to avoid calculating an estimate of marginal likelihood (and therefore running the particle filter) under the MJP for proposals that are likely to be rejected, by implementing a preliminary screening step that uses a cheap approximation of the marginal likelihood. We explored two approximations, the chemical Langevin equation (CLE) and the linear noise approximation (LNA). The LNA can be viewed as an approximation to the CLE. Thus, providing the Euler time-step is not too large the CLE leads to a greater effective sample size over a fixed number of iterations. However under Gaussian observation regimes the marginal likelihood under the LNA is tractable, whereas the marginal likelihood under the CLE is generally intractable whatever the observation regime. We therefore replaced the true posterior under the CLE approximation with a stochastic approximation to this, also obtained via a particle filter. We tested both schemes on a Lotka–

Volterra system where the observed counts follow a Poisson distribution with expectation equal to the true count. We showed how the LNA can be used to obtain a reasonable deterministic approximation to the marginal likelihood even though the observations are not Gaussian and created a scheme which is approximately an order of magnitude more efficient than the standard PMMH scheme. Even though the particle filter is computationally much more costly than simply integrating the LNA, using the CLE we are still able to double the efficiency compared with the standard PMMH scheme. In a further application of the LNA scheme to a more complex MJP, with a larger number of unknown parameters we again obtained a speed up of approximately an order of magnitude.

The proposed methodology can in principle be applied to any stochastic kinetic model and in Sect. 4.3 we applied the delayed acceptance scheme (using the LNA) to a simple epidemic model. For this example, we found that an estimate of marginal likelihood under the MJP could be computed relatively cheaply. In spite of this, running the delayed acceptance scheme is still worthwhile, and we observed an overall increase in efficiency of at least a factor of two.

The efficiency of both proposed delayed acceptance PMMH schemes can be improved in a number of ways. Both schemes can be parallelised and will benefit from recent work on the use of graphics cards for Monte Carlo methods (Lee et al. 2010). In addition, in high signal-to-noise scenarios, the variance of the marginal likelihood estimator under both the CLE and MJP could be reduced through implementation of an auxiliary particle filter such as that considered by Pitt et al. (2012). The interplay between the number of particles, and choice of scaling for the RWM proposal, and the efficiency of the scheme is non-trivial. For example, increasing the number of particles increases the CPU time per iteration but (e.g. Andrieu and Roberts 2009) should lead to a more efficient PMMH algorithm in terms of ESS for a fixed number of iterations. However with a delayed acceptance algorithm we might expect less of an increase in ESS once the accuracy of the stochastic approximation exceeds that of the deterministic approximation since the Stage 2 acceptance rate depends on the ratio of these. Our tuning of the algorithms was relatively ad hoc; with sound tuning advice driven by theory it is possible that further efficiency gains might be obtained.

Our demonstration of detailed balance showed that when a stochastic estimate of the marginal likelihood is used at Stage 1 as well as Stage 2, the independence of the estimators is unnecessary. This suggests that a positive correlation between the two might increase the Stage 2 acceptance rate; unfortunately it was not obvious how to achieve this for our particular estimators. It is also straightforward to extend our derivation to apply to a k-Stage delayed acceptance algorithm, using a sequence of $k-1$ approximations. Such a sequence would need a careful design as the increase in accuracy at each stage would need to outweigh the increase in computational cost, and we do not pursue this here.

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Appendix

Recall that $\mathbf{x} = \{x_t \mid 1 \le t \le T\}$ denotes values of the latent MJP and $\mathbf{y} = \{y_t \mid t = 1, 2, \ldots, T\}$ denotes the collection of (noisy) observations on the MJP at discrete times. In addition, we define $\mathbf{x}_t = \{x_s \mid t-1 < s \le t\}$ and $\mathbf{y}_t = \{y_s \mid s = 1, 2, \ldots, t\}$.

PMMH scheme

The PMMH scheme has the following algorithmic form.

1. Initialisation, $i = 0$,
   (a) set $c^{(0)}$ arbitrarily and
   (b) run an SMC scheme targeting $p(\mathbf{x}|\mathbf{y}, c^{(0)})$, and let $\widehat{p}(\mathbf{y}|c^{(0)})$ denote the marginal likelihood estimate
2. For iteration $i \ge 1$,
   (a) sample $c^* \sim q(\cdot|c^{(i-1)})$,
   (b) run an SMC scheme targeting $p(\mathbf{x}|\mathbf{y}, c^*)$, and let $\widehat{p}(\mathbf{y}|c^*)$ denote the marginal likelihood estimate,
   (c) with probability $\min\{1, A\}$ where
   $$A = \frac{\widehat{p}(\mathbf{y}|c^*)p(c^*)}{\widehat{p}(\mathbf{y}|c^{(i-1)})p(c^{(i-1)})} \times \frac{q(c^{(i-1)}|c^*)}{q(c^*|c^{(i-1)})}$$
   accept a move to c^* otherwise store the current values

Note that the PMMH scheme can be used to sample the joint posterior $p(c, \mathbf{x}|\mathbf{y})$. Essentially, a proposal mechanism of the form $q(c^*|c)\widehat{p}(\mathbf{x}^*|\mathbf{y}, c^*)$, where $\widehat{p}(\mathbf{x}^*|\mathbf{y}, c^*)$ is an SMC approximation of $p(\mathbf{x}^*|\mathbf{y}, c^*)$, is used. The resulting MH acceptance ratio is as above. Full details of the PMMH scheme including a proof establishing that the method leaves the target $p(c, \mathbf{x}|\mathbf{y})$ invariant can be found in Andrieu et al. (2010).

SMC scheme

A sequential Monte Carlo estimate of the marginal likelihood $p(\mathbf{y}|c)$ under the MJP can be constructed using (for example) the bootstrap filter of Gordon et al. (1993). Algorithmically, we perform the following sequence of steps.

1. Initialisation.

(a) Generate a sample of size N, $\{x_1^1, \ldots, x_1^N\}$ from the initial density $p(x_1)$.
(b) Assign each x_1^i a (normalised) weight given by

$$w_1^i = \frac{w_1^{*i}}{\sum_{i=1}^N w_1^{*i}}, \quad \text{where} \quad w_1^{*i} = p(y_1|x_1^i, c)\,.$$

(c) Construct and store the currently available estimate of marginal likelihood,

$$\widehat{p}(y_1|c) = \frac{1}{N}\sum_{i=1}^N w_1^{*i}\,.$$

(d) Resample N times with replacement from $\{x_1^1, \ldots, x_1^N\}$ with probabilities given by $\{w_1^1, \ldots, w_1^N\}$.

2. For times $t = 1, 2, \ldots, T-1$,

(a) For $i = 1, \ldots, N$: draw $\mathbf{X}_{t+1}^i \sim p\big(\mathbf{x}_{t+1}|x_t^i, c\big)$ using the Gillespie algorithm.
(b) Assign each $\mathbf{x}_{t+1}^i$ a (normalised) weight given by

$$w_{t+1}^i = \frac{w_{t+1}^{*i}}{\sum_{i=1}^N w_{t+1}^{*i}}, \quad \text{where} \quad w_{t+1}^{*i} = p(y_{t+1}|x_{t+1}^i, c)\,.$$

(c) Construct and store the currently available estimate of marginal likelihood,

$$\begin{aligned}\widehat{p}(\mathbf{y}_{t+1}|c) &= \widehat{p}(\mathbf{y}_t|c)\widehat{p}(y_{t+1}|\mathbf{y}_t, c)\\ &= \widehat{p}(\mathbf{y}_t|c)\frac{1}{N}\sum_{i=1}^N w_{t+1}^{*i}\,.\end{aligned}$$

(d) Resample N times with replacement from $\{\mathbf{x}_{t+1}^1, \ldots, \mathbf{x}_{t+1}^N\}$ with probabilities given by $\{w_{t+1}^1, \ldots, w_{t+1}^N\}$.

Marginal likelihood under the linear noise approximation

Assume an observation regime of the form

$$Y_t = G'X_t + \varepsilon_t\,, \qquad \varepsilon_t \sim \mathrm{N}(0, \Sigma)$$

where G is a constant matrix of dimension $u \times p$ and ε_t is a length-p Gaussian random vector.

Now suppose that $X_1 \sim N(a, C)$ *a priori*. The marginal likelihood under the LNA, $p_a(\mathbf{y}|c)$ can be obtained as follows.

1. Initialisation. Compute

$$p_a(y_1|c) = \phi\left(y_1\,;\, G'a\,,\, G'CG + \Sigma\right)$$

where $\phi(\cdot\,;\, a\,,\, C)$ denotes the Gaussian density with mean vector a and variance matrix C. The posterior at time $t = 1$ is therefore $X_1|y_1 \sim N(a_1, C_1)$ where

$$\begin{aligned}a_1 &= a + CG\left(G'CG + \Sigma\right)^{-1}\left(y_1 - G'a\right)\\ C_1 &= C - CG\left(G'CG + \Sigma\right)^{-1}G'C\,.\end{aligned}$$

2. For times $t = 1, 2, \ldots, T-1$,

(a) Prior at $t+1$. Initialise the LNA with $z_t = a_t, m_t = 0$ and $V_t = C_t$. Note that this implies $m_s = 0$ for all $s > t$. Therefore, integrate the ODEs (6) and (10) forward to $t+1$ to obtain z_{t+1} and V_{t+1}. Hence

$$X_{t+1}|\mathbf{y}_t \sim N(z_{t+1}, V_{t+1})\,.$$

(b) One step forecast. Using the observation equation, we have that

$$Y_{t+1}|\mathbf{y}_t \sim N\left(G'z_{t+1}, G'V_{t+1}G + \Sigma\right).$$

Compute

$$\begin{aligned}p_a(\mathbf{y}_{t+1}|c) &= p_a(\mathbf{y}_t|c)\,p_a(y_{t+1}|\mathbf{y}_t, c)\\ &= p_a(\mathbf{y}_t|c)\,\phi\left(y_{t+1}\,;\, G'z_{t+1}\,,\, G'V_{t+1}G + \Sigma\right).\end{aligned}$$

(c) Posterior at $t+1$. Combining the distributions in (a) and (b) gives the joint distribution of X_{t+1} and Y_{t+1} (conditional on $\mathbf{y}_t$ and c) as

$$\begin{pmatrix}X_{t+1}\\ Y_{t+1}\end{pmatrix} \sim N\left\{\begin{pmatrix}z_{t+1}\\ G'z_{t+1}\end{pmatrix}, \begin{pmatrix}V_{t+1} & V_{t+1}G\\ G'V_{t+1} & G'V_{t+1}G + \Sigma\end{pmatrix}\right\}$$

and therefore $X_{t+1}|\mathbf{y}_{t+1} \sim N(a_{t+1}, C_{t+1})$ where

$$\begin{aligned}a_{t+1} &= z_{t+1} + V_{t+1}G\left(G'V_{t+1}G + \Sigma\right)^{-1}\left(y_{t+1} - G'z_{t+1}\right)\\ C_{t+1} &= V_{t+1} - V_{t+1}G\left(G'V_{t+1}G + \Sigma\right)^{-1}G'V_{t+1}\,.\end{aligned}$$

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